

BT505 - Biosensors

Final Term notes + Current Questions (2021)

Lesson # (100-203)

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1- DNA Damage:

1. DNA can be **damaged** by **oxidizing agents**, **alkylating agents** and **high-energy electromagnetic radiation** such as **ultraviolet light** and **X-rays**.
2. DNA is damaged due to excessive levels of **Reactive Oxygen Species**. Normal level of ROS may be necessary for **memory and learning**.
3. DNA damaged is related to **cancer** and other diseases.

2- Electrochemical properties of DNA bases

All four bases of DNA can be **oxidized** and produce their **oxidation signals** at **highly positive potentials** on carbon electrodes. The one **electron reduction potentials** of DNA nucleosides are

1. Guanine 1.29
2. Adenine 1.42
3. Cytosine 1.6
4. Thymine 1.7

3- Drawbacks of Label Free Electrochemical Detection

1. It requires **high potential** to detect the Guanine oxidation.
2. **High background signals**.
3. **Non-specific adsorption** of DNA target containing Guanine bases.

4- Features of Glassy Carbon (Allotropic form of Carbon)

1. High-**purity**
2. Excellent **Stability** as high as at **3,000 °C** in vacuum
3. Stable at **500 °C** in the air
4. **Inert** against Chemical **erosion**
5. **Impermeability** to Gas and Solution
6. Significant **Hardness/ Strength**
7. Favorable **electric conducting** property
8. Highly **Resistant** against **Inorganic and Organic salts**
9. Nonporous
10. Good **Bio-Compatibility**

5- Electrochemical Detection HIV target DNA

1. A single-stranded PNA molecule is non-covalently linked the graphene surface with a **pyrene** linker group.
2. The complementary DNA-strand binds with HIV target DNA.
3. 2D graphene with excellent electronic properties help to detect this hybridization event at a very low noise level.

Advantages:

The changes in oxidation and reduction (with and without hybridization) are bases of the detection. Label-free electrochemical DNA hybridization method is

1. Simple
2. Rapid
3. Highly sensitivity
4. Reproducibility
5. Long-term stable.

6- Floropore and Quencher

1. **Fluorophore** is attached at one end of one arm and Quencher is attached at the end of other arm.
2. **Quenching** is a process which decreases the fluorescence intensity of a substance.
3. When the **fluorophore and quencher** are in **close proximity**, the quencher absorbs the energy emitted from an excited fluorophore, thereby **suppressing** its **emission**.

Designing and principle:

Molecular beacons **do not fluoresce** in their **free form**.

In the absence of targets, the **probe is dark**, because the stem places the **fluorophore close to the non-fluorescent quencher**.

Quencher and fluoropore **transiently share electrons**, **eliminating** the **ability** of the fluorophore to **fluoresce**.

Example of Quencher **Cl**, **Acrylamide**, **NaI**.

Working

1. When the **probe encounters a target molecule**, it forms a **probe-target hybrid**.
2. The **hybrid is longer** and more **stable** than the **stem hybrid**.
3. **After hybridization the quencher moved away from the fluorophore.**
4. As the quencher is away from the **fluorophore**, the **fluorescence** of the fluorophore is the restored.

7- Hairpin DNA probe (HDP)

Hairpin DNA probes (HDPs) are **specially designed ssDNA** and have excellent sensing specificity. HDP is immobilized on the electrode surface.

1. Hairpin DNA probe does **not** bind with the **enzyme probe until it hybridizes with the target molecule**.
2. Binding of target molecule with the DNA leads to the generation of signals by the enzyme.

8- Lab On A chip (LOC)

1. A lab-on-a-chip is a device that can **perform the functions of a laboratory on** a single chip of only millimeters to a few square centimeters in size.
2. LOCs deal with the handling of **extremely small liquid volumes** up to Pico liters.
3. Lab-on-a-chip devices are a subset of **micro-electro-mechanical systems (MEMS)** devices.

Development of LOC technology

1. The development of LOC technology is related to **human genome sequence project**, where fast running technologies were established to sequence the whole human genome.
2. Later this approach was used to establish LOC by **Defense Advanced Research Project Agency (DARPA)** of USA for military applications.

Components of LOC technology

For understanding the LOC technology it is require studying the following in detail.

1. Materials for chip
2. Lithography
3. Capillary Electrophoresis
4. Microfluidics
5. Introduction of sample/Reagent onto LOC
6. Mixing of Reagents in LOC
7. Electronic circuits

Advantages:

1. Required less sample volumes for diagnostics
2. Low fluid volumes consumption
3. Less waste
4. Lower reagents costs
5. Fast response
6. Fast Analysis
7. Small heat capacity
8. Fast heating

9- Reynolds Number:

1. Reynolds number is a **dimension less quantity**.
2. Reynolds number **compares** the **effect of the momentum** of a fluid to the **effect of viscosity**.
3. The Reynolds number is the **ratio of inertial forces to viscous forces**.
4. It can predict if a flow condition will be **laminar or turbulent**.

Reynolds Number = Inertial Force / Viscous Force

V = velocity of the fluid

μ = viscosity of fluid

L = length or diameter of the fluid.

10-Photoresist Types

A photoresistor is a type of resistor whose **resistance decreases when the intensity of light increases**.

- **Positive photoresisters:**

On the exposure of light positive photo resistor are dissolved.

- **Negative photo resisters:**

These photoresistor are not dissolved on exposure to light.

Example: Epoxy based polymers.

11-Etching

Etching in micro-fabrication is the **removal of unwanted photoresister** from the wafer. There are two methods of etching.

1- Wet Etching

Hydroflouric acid is used to bath the treated silicon wafers for wet etching.

2- Plasma Etching

Standard conditions for such as pressure under vacuume is created for etching.

12-Freeze drying

This technique is also known as **lyophilization**. It involves the removal of water molecules from the sample by

1. **Freezing the sample**

2. **Lowering the pressure creating a vacuum.**

13-Trypsin stabilization

Trypsin enzyme activity was stabilized in the presence of Trehalose and was immobilized by using two different techniques.

Spray dried immobilization Freeze dried Immobilization.

In each case trehalose act as a stabilizer by forming **hydrogen bond with the enzyme** and the **glass matrix** that was used for immobilization.

14-Sugars as stabilizer

- Sugars in general **protect proteins against dehydration** by hydrogen bonding to the dried protein **by serving as water substitute.**
- Sugars **protect** proteins against loss of activity, **chemical and thermal denaturation.**
- **Trehalose** has also been found to be very effective in the stabilization of labile proteins during **lyophilization** and exposure to **high temperatures** in solution.
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15-Methods for producing Micro-dispenser

There are two methods for producing micro-dispenser.

1. **Cold-forming method:** In which **no heat** is applied and **aluminum** is used as a substrate.

2. **Thermoforming:** It is **manufacturing process**. In this method **plastic polymers** are used and **heat** is applied.

16-Superhydrophobiccoating of Micro channels

A hydrophobic coating is a **thin surface layer** that **repels water**. Droplets of samples hitting super hydrophobic coating can **fully bounce**.

1. It **minimizes** the **surface bio-adhesion** of the blood.

2. The blood **clotting time** is **increased**.

3. **Loss** of Analyte molecules is **prevented**.

17-Microfluids:

The study behavior of fluids at micro level 100 nanoliters to 500 microlitres is known as Microfluids. A number of factors play important role in Microfluids.

1. **Surface tension**

2. **Energy dissipation,**

3. **Fluidic resistance**

Due to flow of blood in capillaries, microfluidics is of great importance.

18-Mixing in LOC

Types of microfluidic mixing:

Controllable and fast mixing techniques are required.

There are two types of mixing strategies.

1. **Passive Mixing** in this approach the **designing of the micro-channels facilitate the mixing** without any external force.
2. **Active mixing.** In this case, mixing efficiency is increased by **external forces applied** to the samples.

19-T Shaped micro-mixer

In T-shaped micro mixer consist of **two inlets** and **one outlet**.

1. The **micro-channels with two inlets** for samples are **flowed perpendicularly** to each other.
2. Mixing **efficiency** can be **increased** by adding some **barriers and obstacles** in mixing channels, which **creates** additional **perturbations**.

20-Properties of Fe based Nano Particles:

Iron based nanoparticles contribute to chemical properties of the NP.

- i. **The zero-valent iron** in the **core** mainly provides the **reducing power** for the reactions.
- ii. **The oxide shell** provides sites for **adsorption**. **Adsorption** also occurs on the **iron oxides surface**, while **Ferric oxide** possesses reducing power.

21-Magnetic Nano Particles as a carrier:

The **core and shell** are the **potential carriers**. The active agents are attached to the **shell** can be adsorbed into the **magnetic core**,

1. The active agents can interact with the shell via **hydrophobic or hydrophilic** interactions.
2. **Active agents** can be **distributed** between the two components of the **core** and the **shell** structures.

22-Use of Super-hydrophobic membrane

Use of super hydrophobic membrane coating could be the first step in the blood sample processing.

1. It **minimizes the surface bio-adhesion** of the blood.
2. The blood clotting **time is increased**.
3. Blood cell **hemolysis is reduced**.
4. Loss of target molecules is **prevented**.

23- Architecture Types on LOCs (Lab-on-a-chip)?

There are different types of architectural design are available for LOC platforms.

1. **Lateral flow platform**
2. **Linear actuated platform**

3. Microfluidic large scale integration platform
4. Centrifuged microfluidics platform
5. Electro-wetting platform

The main components of LOC

1. Injector
2. Transporter
3. Preparator
4. Mixer
5. Reactor
6. Separator
7. Detector
8. Controller
9. Power supply

24-Cell phone-based sensor for detection of *E. coli*

1. A cell phone-based **fluorescent imaging** can detect the presence of *Escherichia coli* in samples.
2. **Antibody** functionalized **glass capillaries** with **quantum dots** can **detect 5 CFU/mL of *E. coli*** in **liquid samples**.
3. The system uses a **lightweight, compact attachment** to cell-phone camera.

25-WHO ASSURED criteria for low cost diagnostics developments?

1. Affordable
2. Sensitive
3. Specific
4. User friendly
5. Rapid and Robust
6. Equipment-free
7. Deliverable to end users

26-Paper Base LOC Impact on flow rate

The movement of fluid through paper is governed by the **physical properties** of the **substrate** and the fluid. **Physical properties of substrate** that can change the flow rate include

- i. **Permeability**
- ii. **Geometry**
- iii. **Evaporation effects.**
- iv. **Pore size**

Impact of physical properties of the fluid of the flow rate

The movement of fluid through paper is governed by the physical **properties of the fluid** include.

1. **Viscosity**
2. **Density**
3. **Contact angle**
4. **Temperature**

5. Surface tension

Flow Control on LOC

There are various ways to control the fluid flow in the channels.

1. Changing the **channel width and length**.
2. Altering the **wettability** of the paper.
3. **Diverting** some fluid through a parallel channel.
4. Changing the **viscosity** of the fluid.

27-Standard Paper-based 3D microfluidic device

The device consists of **2 layers**;

1. The **test solutions are loaded** to injection zone.
2. The **chemical reaction takes place in** folded paper-based 3D microfluidic device.
3. Paper based microfluidic device is then air dried.
4. Image of result is visible after unfolding the paper device.

29- Conductive INK Pen

1. A conductive pen is a **liquid polymer delivery system** a standard ink pen.
2. The polymer is electrically conductive and is rapid to dry.
3. They are used to **create or repair conductive traces** on **printed circuit boards**.
4. **Silver-infused conductive inks** used in modern **transit tickets**.

NP and conductive Ink

1. Most conductive inks are **based on metal NPs**.
2. NPs are **inserted into a solvent** at a certain loading ratio to provide the **wanted viscosity**.
3. It results in a liquid solution, called the “the metal ink,” with the **NPs in suspension**.
4. Average size of the particle varies between **10 and 100 nm**.

30- Working of Paper Battery

The working principal of paper battery is same as for the conventional battery.

1. **Battery is dipped in ion-based liquid**, then the battery starts working.
2. There is a **chemical reaction** between the electrodes and liquid.
3. **Electricity is generated** by the movement of electrons from cathode terminal to anode terminal.

31- ARRAY

An array is a **data structure** that contains a **group of elements**.

The collection **of Elements** has

- Same Name**
- Same Data type**

1. Arrays are arranged in **natural numbers** (any integer greater than zero).
2. **In Arrays data is organized** that a **related set of values** can be **easily sorted or searched**.

Features of Arrays

A **key characteristic** of this system in Biosensor is its **two-dimensional configuration**. It allows **controls** and **standards** to be analyzed in **parallel with unknowns**.

Array biosensor is **capable of**

1. Multiple samples simultaneously
2. Rapid Testing
3. on site screening
4. Reliable Detection

32- Biosensor Arrays in an Electrophysiology

Organoids are

- 3-D assemblies of cells produced by tissue culture
- Containing more than one cell type
- Exhibit at least some physiological features of an organ.

A 3-D micro sensor array can be fabricated to study electrophysiology **of heart**.

Optical Sensor Array

The human visual system is more sensitive to green color than others. Human eye can differentiate **green color from ten million shades**

1. The **increased density of the green sensor improves** the optical signal.
2. Optical sensor with **more green color sensor** can provide better **resolution** for the eye.

Sensor Array in Camera

A sensor array is a group of sensors, usually **deployed in a geometry pattern**, used for **collecting and processing** signals. **In a camera**, the data from the three types of **color** sensors is used to form an image.

1. Red (R)

2. Green (G)

3. Blue (B)

A specific design for (R, G, B) sensors is used.

33- Why newspapers become yellow after some time?

The exposure of lignin to air and sunlight for some time turns paper yellow.

In the presence of **light and oxygen**, **lignin reacts to give yellow** color to the paper. That is why **newspaper becomes yellow after some time**.

Types of paper:

In market there are more than 10,000 types of papers are available, depending upon the composition and end use.

1. Filter paper

2. Chromatographic paper

3. Carbon paper

4. Printing paper

5. Nitrocellulose paper

6. Copy paper

7. Wax paper

34- Write a note on network in paper structure?

Network in paper structure:

Paper has a Network in its structure;

1. **Fibrous** (cellulose) with inter and intra chain hydrogen bonds.
2. **Non-fibrous** (fillers) materials.
3. **Capillaries**. With a variety of capillary system.
4. **Pores**. Paper structure has a complicated set of cavity pore.

35- Give the mechanism of fluid transport in paper micropad?

Transport of fluid on Paper:

When a **droplet of liquid** contacts with paper, the **competition of 4 main phenomena**

1. Droplet wets the surface of paper
2. Paper absorbs and wicks the droplet by capillary action
3. Evaporation proceeds
4. Diffusion occurs.

All these four steps lead equilibrium conditions.

36- Name the strategies that are used to program the paper based micro fluids. Different strategies are employed for effective fluid flow control on paper is adopted?

1. **Geometry Based methods:** It involves the **variations in geometrical designs**.
2. **Chemical based methods:** requires the applications of chemicals.
3. **Physical based methods:** are involved in **physical forces** are regulated.

37- Name two chemicals that are used in controlling the fluid flow in paper based micro pads?

1. **Wax:** It is hydrophobic organic molecule that is used for surface coating in micro fluids.
2. **Surfactants:** Surfactants are compounds that lower the surface tension (or interfacial tension) between two liquids, between a gas and a liquid, or between a liquid and a solid

38- Name three types of sensors that could be integrated on paper?

Different types of sensors can be integrated successfully, on paper microfluidics

1. Chromogenic sensors
2. Electrochemical sensor
3. Humidity sensors
4. Optical sensor
5. Pressure sensor
6. PH sensor
7. Proximity Sensors
8. Thermal sensor

39- What is the advantage of white color of filter paper for optical sensor?

The color of filter paper is **white** that's why **any change** in the color during the detection of analyte is easily detected.

40- Give the principle for the development of humidity sensor on a paper?

Fabrication and Principle of Humidity sensor:

1. On the paper electrodes **are directly drawn onto the paper** using conductive ink.
2. The porous structure of the paper readily absorbs moisture from the air.
3. There is a marked **change in the capacitance of the sensor** due to the **adsorption of hydroxyl groups** from the water onto the paper.

41- How temperature sensor could be setup on a paper?

The temperature sensors are made out of a simple resistive structure using metals.

1. The principle of operation is based on the change in resistance of the metal
2. Aluminum foil or Silver ink is used for drawing circuits.
3. Change in the .temperature leads to a variation in resistance that is recorded as a signal.

42- Applications of Paper micro fluids?

Paper based microfluidic devices have **advanced significantly**, because they are

1. **Affordable**
2. **Automated Flow**
3. **Portable**
4. **Biodegradable diagnostic platforms**

They have variety of applications in **three main sectors health, environmental, and food quality** applications

43- 2 ways of Integration of Nanostructures?

The function of Nanomaterial is to improve **the sensing mechanisms**. Nanomaterial can **be integrated in 2 ways**.

1. The nanostructure act as an intermediate layer between biological agents and physicochemical components.
2. Biological agents, and the transducer are combined with nanomaterial.

44- Silk Fiber as protective Protein?

SF is a **natural protein** that is extracted from silkworm cocoon consists of

1. **Sericin protein (Outer coating)**
2. **Fibroin protein (Central core).**

Fibroin is largely made up of the amino acids **Gly-Ser-Gly-Ala-Gly-Ala** and forms **beta pleated sheets**. There is a network of inter and intra-chain hydrogen bonds in protein structure.

Silkworm silk sericin

1. Sericin is composed of 18 different Amino Acid residues, of which **32% is serine**.
2. It is a hydrophilic glycoprotein.
3. Sericin is an amorphous protein.
4. Sericin, consists of random hydrophobic and hydrophilic blocks of amino acids,
5. It contains
 - **Flavonoids**
 - **Carotenoids**

That act as gumming agents.

45- Silk Fibroin as protective Protein?

A schematic illustration of silk fibers produced by silkworms:

- (A) The raw silk fiber is composed of two fibroin fibers held together with sericin covered with a protein coat.
- (B) The illustration of β -sheet crystallite embedded in the amorphous matrix of silk fibroin fibers
- (C) Each silk fibroin heavy chain consists of hydrophobic and hydrophilic repetitive domains (GAGASA).

46- Superoxide Dismutase

- SOD is an antioxidant enzymes (EC 1.15.1.1) is an enzyme.
- **Superoxide dismutase (SOD)** is the first **detoxification enzyme** and most powerful **antioxidant in the cell**.
- It is an important endogenous antioxidant enzyme that acts as a component of first line defense system against reactive oxygen species (ROS).
- **Superoxide** is produced as a **by-product of oxygen metabolism**.

Superoxide Dismutase Families:

There are three major families of superoxide dismutase, depending on the protein fold and the metal cofactor:

1. **Cu/Zn type SOD**: The enzyme binds both copper and zinc, for its functioning.
2. **Fe and Mn types SOD**: The enzyme bind either iron or manganese,
3. **Ni type SOD enzyme** type which binds nickel, for its activity.

Cerium and its Uses:

1. Cerium has a variable electronic structure.

2. The energy of the 4f electron is nearly the same as that of the outer 5d and 6s electrons.
3. These electrons are delocalized in the metallic state, and only a small amount of energy is required to change the relative occupancy of these electronic levels.
4. Cerium has dual valence states that are +3 and +4.

47- Quantum Dots:

- Quantum Dots are made up of **semiconductor**.
- They are nano-sized **crystalline structures (ranging from 2 to 10 nm)**.
- Qds possess **superior fluorescent** properties with less photo bleaching.
- **Photo bleaching** (sometimes termed **fading**) is the photochemical alteration in a **fluorophore molecule**.

Quantum Dots and wave length:

- The biggest quantum dots produce the longest wavelengths and lowest frequencies.
- The smallest dots make shorter wavelengths and higher frequencies.
- Big dots make red light and small dots make blue, with intermediate-sized dots producing green lights.

Uses

1. Quantum dots are also being used in place of Organic Dyes in biological research,
2. They can be used like nanoscopic light bulbs to light up and color specific cells that need to be studied under a microscope.
3. Qds being tested as sensors for chemical and biological warfare agents such as anthrax.

48- What are the properties of nano-diamond?

Properties:

A distinct combination of outstanding mechanical performance with

1. Biocompatibility
2. Chemical resistance
3. Doping can be induced
4. Hardest of the known materials
5. Electronic properties
6. Highest atomic density.
7. Stable in acidic /basic environments
8. Magneto-optical properties

49- What is Nitrogen vacancy and how it improves the functional properties of nano-diamond?

Nitrogen vacancy:

1. Occasionally a nitrogen atom has substituted for one of the carbon, known as Nitrogen vacancy.
2. It improves the optical and magnetic properties of NDs.
3. A nitrogen atom has five valence electrons.
4. Three of them covalently bond to the carbon atoms and two remain non-bonded and are called a lone pair.
5. It improves the performance of NDs.

50- What is meant by the core-shell nanowires? Give at least two examples of core-shell nanowires?

Core/Shell Nanowires have

1. Core

2. Shell

Core and shell are made up of different materials, can consist of different elements.

1. They have large surface area.
2. Core/shell improves the efficiency of the nanowires.

Examples:

Core/shell NWs consisting of

1. Semiconductor/semiconductor
2. Semiconductor/metal
3. Metal/Metal oxide
4. Metal oxide/Metal oxide
5. Metal/Metal hetero-structure(s).

51- What are properties of core-shell nanowires?

- 1. Doping.** The transport properties can be tuned by means of doping.
- 2. Introduction of Hetrostructures:** By making use of hetero-structures the nanowire properties can be tailored.

3. Use of higher band gap material: By covering the NW radially by a higher band gap material surface, can improve the performance.

The band gap is the energy difference between the top of the valence band and the bottom of the conduction band.

52- Name the nanomaterial's that are used in DNA nanosensors?

Nanomaterial used:

They make use of nanomaterial as transducers.

1. Semiconductor nanoparticles Si
2. Quantum dots (QD)
3. Silver
4. Per fluorocarbon
5. Organic polymers
6. Fluorescent lanthanide nanoparticles

53- Give the properties of PFC?

Per fluorocarbons (PFCs) are

1. Inert synthetic
2. Nontoxic materials.
3. Perfluoro alkanes do not bio accumulate.
4. When used in medical procedures are rapidly excreted from the body, primarily via expiration.
5. PFC is not flammable.

54- What is the importance of optical nanobiosensors?

1. Optical Nanosensors are known as PEBBLES
2. Optical Nanosensors are used in cells to determine physiological functions.

3. The fluorescent of nanomaterial is used for making quantitative measurements.
4. Sensors can be inserted into living cells with a minimum of physical perturbation.

55- Give the fabrication methods for optical nanobiosensors?

PEBBLEs are **biocompatible**, with **negligible biological effects**. Sensor response times are less **than 1 ms**. The sensors are fabricated in a microemulsion.

PEBBLE consists of;

1. Fluorescent indicator dye
2. Polyacrylamide matrix.
3. Reference dyes in the matrix.

PEBBLES:

PEBBLES are spherical optical Nano sensors. Size range is 20 and 200 nm in diameter.

1. **P** PROBES
2. **E** ENCAPSULATED
3. **B** BY
4. **B** BIOLOGICAL
5. **L** LOCALIZED
6. **E** EMBEDDINGS

56- What is an electrochemical nanosensor?

1. Electrochemical sensors **are the most commonly used and widely accepted sensor** in the naobiosensors.
2. The sensors functions **on the principle of electrochemistry**.
3. The electron consumed or generated during the process of bio interaction.
4. The electrochemical signals l produced are measured **by electrochemical methods**.

57- What is the classification of electrochemical nano biosensor?

Classification is based on their working principle; electrochemical nanosensorsdevice could be broadly categorized as

1. Amperometric Nanobiosensors
2. Voltammetric Nanobiosensors
3. Potentiometric Nanobiosensors

58- Give two common examples of electrochemical nano biosensors?

Examples: A e-nose or e-tongue

- 1- An e-nose or e-tongue is combination of gas sensors or chemical sensors which mimics human nose or human tongue.
- 2- Gas sensor arrays are defined as **‘electronic nose (e-nose),’** while chemical sensor arrays are referred to as **‘electronic tongue (e-tongue)’**.
- 3- Rapid sensing can be achieved by those sensor arrays,
- 4- The price is lower than the standard analytical equipment.

59- Human Olfactory system and E-Nose

1. The human olfactory system contains some 100 million olfactory receptors.
2. Nose has approximately 350 different types of membrane receptor binding proteins.
3. The proteins are distributed along the olfactory epithelium (the lining of the nose and part of the olfactory mucosa).

4. In contrast e-nose systems typically have 32 or less chemical sensors in a basic chamber.

60- Give the chemical basis for the detection of caffeine in a given sample by using an electronic tongue?

Detection of Caffiene (C₈H₁₀N₄O₂):

In E-Tongue Caffiene can be detected on the basis of it chemistry. Caffeine is a central nervous system (CNS) stimulant. Caffiene has two functional groups.

- **Amide functional Group**
- **Amine functional groups.**

Caffeine can interact with

- 1. Alkane-like regions**
- 2. Amine**
- 3. Phosphoric groups**

Electronic Tongue Advantages:

Mimicking human taste is **advantageous in situations when human expert panels cannot or should not be applied**, due to

- 1. Process conditions**—as in case of automatic process control especially on an industrial scale.
- 2. Poisonous/extreme**condition samples—e.g., repetitive tasting of drugs and pharmaceuticals.
- 3. Economic reasons**, defined in terms of time or financial expenses.

61- Name different sources for contamination of crops?

A huge amount of pesticides and fertilizers are used in the crops to enhance the crop production and control pest. The remains of pesticides and fertilizers contaminate the crops and are hazardous to human and other organisms.

1. Pesticides, including herbicides, insecticides, and fungicides, Organophosphate

2. Fertilizers such as Urea, metabolites of fertilizers

Biosensors for Agriculture:

1. Testing for pesticides in the crop
2. Soil pH testing
3. Crop deterioration test
4. Hormone detector

Trending Biosensors:

1. E-nose
2. E-tongue

Detection of **odd volatile compounds**, Food process industry, flavor testing

62- What is the role of ethylene in plant physiology?

The monitoring of the freshness of **climacteric fruits is very important**. It is monitored by **measuring plant hormones**. **Ethylene is a gas that serves as a hormone** in plants.

It plays a critical role in plant growth and development such as

1. Stimulating of fruit ripening
2. Regulating the ripening of fruit

3. Opening of flowers,
4. Shedding of leaves.

63- Give the physical properties of Ethylene gas?

Ethylene is a colorless flammable gas with a faint "sweet and musky" odour when pure.

1. Ethylene is a hormone **that affects the ripening and flowering of many plants.**
2. It regulates the ripening and aging of plants.
3. Ethylene is normally produced in small quantities by most fruits and vegetables.

64- Name different preservatives that are used in fruit juices?

Benzoic acid is commonly added as a food preservative in the form of its salts

1. Sodium benzoate (E211)
2. Potassium benzoate (E212)
3. Calcium benzoate (E213)
4. Ascorbic acid / vitamin C (E300)
5. Erythorbic acid (E315)

65- Name the microbe that can produce benzene degrading enzyme?

Pseudomonas putida:

1. Pseudomonas putida is a Gram-negative bacterium that normally lives in water.
2. Pseudomonas putida is a rod-shaped, flagellated.
3. It is a Gram-negative bacterium is commonly used in environmental studies because it is able to degrade many aromatic compounds.

66- Give the reasons for the presence of Antibiotics in the milk sample?

Presence of Antibiotics could be due to two reasons.

1. Contamination from the cow, who is being treated for a disease.
2. Antibiotic is used to increase the life of the Milk.
 - Antibiotics in milk are **relatively stable** to relatively **high temperatures**
 - Low temperatures (0–10°F.).
 - Under refrigeration temperatures milk is stable for several days.

67- Why it is not recommended to use antibiotics to increase the life of the raw milk?

Effect of Antibiotics in Milk:

1. Antibiotic residues can affect people's health and can develop **antimicrobial resistance**.
2. **Penicillin** in very small concentrations found in milk may **cause reactions in highly sensitive individuals**
3. The presence of antibiotic in the milk in sufficient concentrations to **inhibit dairy starter microorganisms**.
4. It causes **economic losses** to the cheese and fermented milk industries.

68- Name the bacterial strain that is used for the detection of beta lactam antibiotics?

For the detection of drug, a bacterial strain *Pseudomonas aeruginosa* is selected.
It is a common

1. **Encapsulated,**
2. **Gram-negative,**
3. **Rod-shaped bacterium**

It can cause disease in plants and animals, including humans.

P. aeruginosa displays resistance to a variety of antibiotics, including β -lactams.

69- What are the structure and functions of Vitamin B12? What happen to body when its deficiency is occurring?

Vitamin B12:

1. The structure of B12 is based on a **corrin ring**
2. The central metal ion is cobalt.
3. There are six coordination sites.
4. Four coordination sites are provided by the corrin ring,
5. **One coordination site** is provided by a **dimethyl-benzimidazole group**.

Functions and Deficiency:

Vitamin B12 performs a key role in the synthesis is of DNA, neurolipids, and cellular growth. It.'s deficiency causes.

1. Anemia
2. Confusion
3. Depression
4. Problems with balance
5. Poor memory
6. Soreness of the mouth or tongue

7. Tiredness

70- Name at least three approaches used for the fabrication of the B12 Biosensors?

Three approaches are used for the fabrication of a Biosensor for the detection and estimation of Vitamin B12 in the samples.

1. Microbial cell Biosensor
2. Intrinsic factor (B12 binding protein Biosensor)
3. Nanostructures coated with D-phenyl Alanine

71- What are the three main constituents of a fertilizer?

Important Nutrients in fertilizers:

1. **Nitrogen (N)**—nitrogen is largely responsible for the **growth of leaves** on the plant.
2. **Phosphorus (P)** —Phosphorus is largely responsible **for root growth and flower and fruit development**.
3. **Potassium (K)** —Potassium is a nutrient that helps the **overall functions** of the plant perform correctly.

72- Give the mechanism of Action of Urease enzyme in the catalysis of urea?

Urease Enzyme:

1. Ureases (EC 3.5.1.5), functionally, is a hydrolase.
2. jack-bean urease 840 amino acids per molecule, of which 90 are cysteine residues.
3. The optimum pH is 7.4 and optimum temperature is 60 °C.
4. Substrates include urea and hydroxyl urea.

5. Lead salts are inhibiting.
6. It is a metallo enzyme requires Ni ions for activity.

73- Give the effects of Organophosphates on human?

Organophosphates:

1. Organophosphates (OP) are chemical substances that are produced by the process of esterification between phosphoric acid and alcohol.
2. These chemicals are the main components of herbicides, pesticides, and insecticides.
3. OPs are also the main components of nerve gas.
4. Acute or chronic exposure to OPs can produce varying levels of toxicity in humans, animals, plants, and insects.

74- Biosensing approach for pesticide detection in Food

There are two approaches that are widely used for the fabrication of biosensor to detect pesticides, based on organo-phosphates.

1. **Enzyme based biosensor** using acetylcholine esterase as bioreceptor element; the inhibition of enzyme is considering a signal for the presence of pesticide.
2. **Microbial cell biosensor.** Microbes capable of hydrolyzing OPH are used as Bio receptor element.

75- Name different food pathogen?

Biosensor detection has not been limited to the common pathogens like

1. Escherichia coli O157:H7,
2. Salmonella,

3. *Listeria monocytogenes*,
4. *Clostridium perfringens*
5. *Staphylococcus*
6. *Campylobacter*,
7. *Bacillus*
8. *Shigella*

76- How viral DNA can be used for the detection of food biosensors?

Bacterial bacteriophage for pathogen detection:

1. Bacterial Bacteriophages bring new quality and opportunities in bacterial detection.
2. Amplification of phages and release of bacterial content can be analytical signals.
3. Phages can be deposited on solid substrates to create sensing layers in biosensors.
4. Phages can serve as biorecognition element.

77- How will you detect more the one pathogenic sample in a food sample?

The detection of the food-borne pathogen-specific genes on thin-film biosensor chips:

1. DNA oligonucleotides sequences for the selected genes from eleven target microbes (food borne pathogens), are used as a bioreceptor element.
2. The probes were spotted by a computer-controlled dispenser.

3. Each spot comprised 40 nlof1 probe solution. Capture probes are printed on the chip.

78- Heavy Metals:

A solid material which is typically hard, shiny, malleable, fusible, and ductile, with good electrical and thermal conductivity

Around 95 of the 118 elements in the periodic table are metals.

There are three categories of metals, as per periodic table.

1. Alkali metals,
2. Alkaline earth metals
3. Transition metals.

79- Explain how heavy metal ions get into the crops?

Transfer of Heavy metals in crops

Heavy metals are transferred from soil pores to plants in ionic forms, which can vary by metal. The bio-speciation of heavy metals can also vary by food crop.

1. Iceberg lettuce
2. Radishes
3. Beans
4. Tomatoes

All accumulate heavy metals with different concentrations in the roots, leaves, and fruits.

Damage caused by Heavy metals

1. Breaks in DNA strands,
2. Molecular cross linkage,
3. Mutations of genetic materials,
4. Oxidative stress and damage by ROS and free radicals,
5. Functional and structural membrane disintegration, which all ultimately increase the phyto availability of heavy metals and inhibit the growth of crop plants

Chromium ions as contaminants

Chromium is a source of environmental pollution and groundwater contamination due to

1. Wrongful discharges,
2. Accidental spills of industrial wastewaters
3. Leakage of leachates from unsuitable landfills.

Cr mostly found in two oxidation states:

- Trivalent [Cr(III)] or
- Hexavalent chromium [Cr (VI)], is a highly toxic mobile oxidant.

80- Why cyanide is known as a poison?

Cyanide is a rapidly acting substance that is traditionally known as a poison.

Hydrogen cyanide is a gas and, potassium cyanide is a salt.

Large numbers of people are daily exposed to low concentrations of cyanogenic compounds.

1. Collapse
2. Relaxant effect on heart & muscle
3. Sedative
4. Vertigo
5. Vomiting

81- What is the inhibitory mechanism of Cyanide on respiratory enzymes?

Mechanism of Action:

It involves measuring the inhibition of respiration.

1. Cyanide interferes with the proper functioning of cytochrome C in the respiratory pathway.
2. As a result, less oxygen is consumed and less carbon dioxide is produced by bacteria in the presence of cyanide.
3. Other compounds such as pesticides and herbicides also decrease the amount of bacterial respiration.

82- Name the bacterial strain that can metabolize cyanide, justify it?

1. Pseudomonas fluorescence strain can use cyanide as its sole nitrogen source.
2. When this strain biodegrades cyanide it produces cyanate, which is then broken down to ammonia and carbon dioxide.
3. The biosensors measures the byproducts of the reactions involved with the degradation of cyanide.
4. This method is more selective

83- Name different foods having anti-oxidant activity?

1. Dark Chocolate
2. Carrots
3. Potatos
4. tomatoes
5. Blueberries
6. Strawberries
7. Artichoke
8. Goji Berries
9. Raspberries
10. Grapes

Health benefits of Antioxidant

Antioxidants are well-known for their beneficial health effects by providing protection against excessive free radical accumulation. Excessive free radical associated with

1. Aging
2. Cancer
3. Neurodegenerative diseases
4. Pulmonary diseases
5. Heart Diseases

84- What is intelligent packaging, give its features?

Intelligent packaging:

Active packaging usually means having active functions beyond the inert passive containment and protection of the product. Intelligent and smart packaging usually involves the ability to sense or measure.

1. The inner atmosphere of the package
2. The Environment

Functional packaging:

Active packaging Incorporation of certain components into packaging systems. In this case ethylene scavengers are used to control the aging process of fruit, inside the packaging.

The functional Packaging can

1. Prolong shelf life
2. Sustain the quality
3. Safety and
4. Retain the Characteristics of the food

85- What is meant by intelligent label?

Intelligent Labels:

It can provide quality characteristics of the food. Examples of these types are

1. Gas leakage Indicator
2. Humidity sensors
3. Temperature indicators
4. Time Indicator

➤ Depending on the monitoring factor.

These systems can be placed outside or inside the package.

86- What is meant by RFID system?

Radio-frequency identification (RFID)-based sensors are used in the smart packaging to detect changes in the physical stimuli. Such as change in

- 1. Humidity**
- 2. Temperature**
- 3. PH**
- 4. Oxygen**

Taken place in the microenvironment in the food package.

An RFID system includes

1. A reader (or interrogator)
2. A transponder (or a tag)
3. Associated antennas

Working of RFID system:

A reader emits radio waves to capture data from an RFID tag,
Data is then passed onto a host computer for analysis and decision making.

87- Name different Chemical warfare agents?

Chemicals of warfare:

These are the chemicals of warfare.

1. MUSTARD GAS
2. NERVE GAS
3. CHLORINE
4. LEWISITE
5. PHOSGENE
6. CYANOGEN CHLORIDE
7. HYDROGEN CYANIDE

88- What is mustard gas? Why it is called mustard gas?

Mustard gas, or sulfur mustard, is a chemical agent that **causes severe burning** of the skin, eyes and respiratory tract. It can be absorbed into the body through

1. Inhalation
2. Ingestion
3. Skin
4. Eyes.

89- What is nerve gas?

Nerve agents, sometimes also called nerve gases, are a class of organic chemicals that **disrupt the mechanisms by which nerves transfer messages to organs**.

The disruption is caused by the blocking of **acetylcholinesterase**.

The enzyme that catalyzes the breakdown of acetylcholine, a neuro-transmitter.

90- Give the role of sulfur compounds in the acid rain and smog?

1. SO₂ concentration of 500 µg/m³ should not be exceeded over average periods of 10 minutes' duration.
2. When SO₂ combines with water, it forms sulfuric acid. It has devastating effect on environment
3. Sulfur di oxide is the main component of acid rain which is a cause of deforestation.

91- How ozone is produced in Biosphere?

1. Ozone or trioxygen, is an inorganic molecule (O₃).

2. It is a pale blue gas with a distinctively pungent smell.
3. It is an allotrope of oxygen that is much less stable than the diatomic allotrope Oxygen breaking down in the lower atmosphere to dioxygen.
4. Ozone at ground level is not to be confused with the ozone layer in the upper atmosphere.

92- When the first DNAzyme was discovered?

1. The first known deoxyribozyme was a ribonuclease, **discovered in 1994 by Ronald Breaker.**
2. Ronald was working as a postdoctoral fellow in the **laboratory of Gerald Joyce** at the Scripps Research Institute.
3. This deoxyribozyme, later named **GR-5.**

Secondary structures of E17 and GR-5

The top green strands are the substrates and these two DNAzymes have the same substrate sequence.

1. The arrow heads point at the cleavage junction.
2. The important nucleotides are numbered.
3. The blue nucleotides are highly conserved.
4. The four red nucleotides in 17E are important.

Advantages

1. DNAzymes have found practical use in metal biosensors.
2. the most commonly used DNAzyme in biosensors is hemin/G-quadruplex DNAzyme.
3. A DNAzyme based biosensor for Pb ion is used to detect lead ion.

DNAzyme shave two advantages over ribozymes:

- I. Stability secondary to containing DNA instead of RNA
- II. An inexpensive cost of production.

93- Give different types of water pollution?

Types of water Pollution:

Most Common types of water pollution include.

1. Agriculture
2. Oil Pollution
3. Plastics
4. Radioactive Water
5. Sewage
6. Urban Development
7. Industrial Wastewater

94- What is Eutrophication?

Eutrophication:

The process by which a **body of water** acquires a **high concentration of nutrients**, especially **phosphates and nitrates**. It promotes excessive growth of algae.

95- What is Anoxia?

It is a lack of **O₂**, caused by excessive nutrients in waterways which trigger algae growth. High levels of organic matter and the **decomposing organisms** deplete the available Oxygen, causing the death of organisms, which turns green or milky white and gives off a strong rotten egg odour.

96- What is meant by Grey water and Black water?

1. Greywater:

Water coming from domestic equipment other than toilets (e.g., bathtubs, showers, sinks, washing machines) is called greywater.

2. Black water:

Black water contains human wastes. Black water is the mixture of

- **Urine**
- **Feces**
- **Flush water**
- **Pathogens**

97- Deicing:

De-icing can be accomplished by mechanical methods (scraping, pushing); through the application of heat; by use of dry or liquid chemicals designed to lower the freezing point of water such as;

1. Salts

2. Brines

3. Alcohols

4. Glycols

by a combination of these different techniques.

98- BIOX-1010:

1. BIOX-1010 is **an on-line analyzer** for continuous BOD measurement in industrial and municipal waste-water treatment plants.

2. BOD data **are generated with a lag time of only 3 to 15 minutes**, with a measurement range of **20 to 1500mg/liter** BOD.
3. Inside the **bioreactor**, **microorganisms grow inside small plastic cylinders** where they are **protected against mechanical abrasion**

99- Name the two agencies in Pakistan responsible for maintain the water quality?

- 1. Pakistan Environmental Protection Agency**
- 2 .National Standards for Drinking Water Quality**

100- What are coliform bacteria?

Coliform commonly known as” indicator organisms”, refers to a wide groups of bacteria (about 16 species) that found throughout the environment and used to evaluate the water quality.

These organisms have a wide distribution in soil, water surface, and vegetation as well as on the skin or intestinal tract of worm-blooded organisms including humans.

Classification:

Coliforms are divided into three main groups. Each group is an indicator of drinking water quality with a different level of risk. Groups are including:

1. Total Coliform Bacteria
2. Fecal Coliform Bacteria
3. E. coli Coliform Bacteria

101- Intimin:

Intimin is an outer membrane bound bacterial protein.

It has a molecular weight of 94KD.

35KD of the protein is composed of virulent part.

It has a major role in causing the infection by binding with the micro villi of the intestine.

Working of intimin:

- Intimin is expressed on the bacterial surface where it binds with the receptor.
- Receptor of intimin is called translocated intimin receptor.
- TIR along with other 25 proteins is secreted for attaching and effacing E.coli cells directly onto the surface of cytoplasm of the epithelial intestinal cells.

102- What are endocrine hormones? Name three endocrine glands?

The endocrine system is comprised of ductless glands that produce hormones that Regulate

1. Metabolism
2. Growth
3. Development
4. Tissue function
5. Sexual function
6. Reproduction
7. Sleep

Examples of Endocrine Glands:

1. Pituitary gland
2. Thyroid gland

3. Pineal gland
4. Adrenal gland
5. Ovary and
6. Testis

103- Endocrine Disrupter

Endocrine disruptors are chemicals that act **on the endocrine system** of humans and animals by mimicking the hormone blocking and/or Interfering in some manner with the **natural instructions** of hormones to cells and disrupt

- **Developmental**
- **Reproductive**
- **Brain**
- **Immune** and other functions.

Examples:

Some of the major chemicals that are known as **water contaminants** are endocrine disrupting chemicals such as **disinfection byproducts, fluorinated** substances.

1. **Bisphenols**
2. **Phthalates**
3. **Pesticides**
4. **Natural and synthetic estrogens**

Fluorinated substances also can be found in drinking water.

104- What are the important factors required for designing a diagnostic Biosensor?

Diagnostic biosensor is used for **Diagnosis of disease, monitor diseases, drug discovery, and detection of disease causing pathogens.**

In the fabrication of Biosensors there are five main points that are focused.

1. **Nature of Analyte or target molecule**
2. **Sample matrix**
3. **Types of Bio recognition element**
4. **Transduction process**
5. **Material/Method for immobilization**

105- What are lipocalins, give their function?

Protein Lipocalins:

The lipocalins are a family of proteins which **transport small hydrophobic molecules** such as

1. Steroids
2. Bile pigments
3. Retinoids
4. Lipids.

They share limited regions of sequence homology and common tertiary structure architecture.

106- What is gravimetric analysis? What are gravimetric biosensors?

Gravimetric analysis is a **class of lab techniques** used to **determine the Mass and Concentration of a substance** by measuring a **change in mass**.

1. The first **gravimetric biosensor** was a **quartz crystal microbalance (QCM)**.
2. It was based immunosensor for protein detection, it was reported in 1972.

107- FET for the detection of EBOLA Biosensors?

1. Schematic diagram of the rGO-based FET biosensor.
2. rGO sheet was deposited onto the device to bridge the drain and the source electrode.
3. Al₂O₃ was coated on the rGOs heet for surface passivation.
4. Ebola antibodies were conjugated with gold NPs on the channel and functioned as sensing probes for capturing Ebola antigens.

Conductometric Biosensors:

Conductometric biosensors have important advantages:

1. They **do not need the use of a reference electrode**;
2. They are **insensitive to light**.
3. They can be miniaturized.
4. They can be **integrated easily using a cheap thin-film standard technology**.

Potentiometric Diagnostic Biosensor:

1. Potentiometric Biosensors allow label-free detection.
2. Sweat is a **biological fluid**. Sweat analysis is noninvasive.

3. **Sweat** is the most popular sample substrate for wearable sensors.

4. The sweat composition is linked to various physiological processes that makes it suitable for the diagnostics of health status and sports performance.

Impedimetric Biosensors:

1. Impedimetric biosensors **offer label free detection** of target molecules.
2. They can detect small molecules, large molecules and even cells.
3. It is like resistance, but it also takes into account the effects of capacitance and inductance.
4. Impedance is a measure of the overall opposition of a circuit to current,
5. Impedance is measured in ohms

108- Surface Plasmon Resonance imaging:

1. Surface Plasmon Resonance imaging (SPRi) is a label-free optical detection to monitor and analyze bio molecular interactions in real time. .
2. It analyzes bio molecular interactions in **real time**.
3. The imaging capability enables users to visualize the
 - **Entire working area** and
 - **Work in a multiplex format.**

109- What is meant by LAMP?

1. Loop-mediated isothermal amplification technique is used them for DNA detection.

2. LAMP is efficient detection method that **amplifies DNAs** using **DNA polymerase**.
3. LAMP is carried out in **isothermal conditions**.
4. The LAMP results are highly **repeatable, accurate, and highly selective** for the target DNA sequence.

110- Name different dyes that could be used in LAMP assay?

For naked eye to observe change of colors one can use different dyes

1. SYBR Green I,
2. Hydroxynaphthol blue (HNB)
3. Propidium iodide.
4. It also can be detected by color change caused by **gold nanoparticles** (AuNP) probes.

111- Ferromagnetism and Ferrimagnetism:

Ferromagnetism:

It is the basic mechanism by which certain materials (such as iron) form permanent magnets, or are attracted to magnets.

Ferrimagnetism:

It is a type of **permanent magnetism that occurs in solids** in which individual atoms spontaneously align themselves,

1. Parallel
2. Others generally antiparallel,
3. Paired off in opposite directions

112- Paramagnetism:

Paramagnetism is a form of magnetism.

1. Some materials are weakly attracted by an externally applied magnetic field.
2. Paramagnetic properties are due to the presence of some unpaired electrons.
3. It from the realignment of the electron paths caused by the external magnetic field.

113- Super paramagnetism:

1. **Superparamagnetism** is (physics) paramagnetism that occurs below the Curie temperature.
2. **Curie temperature** (TC), or Curie point, is the temperature above which certain materials lose their permanent magnetic properties, which can (in most cases) be replaced by induced magnetism.

114- Magnetoresistance:

Magneto resistance is the change in the resistance of a material due to the application of a magnetic field.

If alternating layers of

1. Ferromagnetic and

2. Non-magnetic metals are used,

This change is much greater than expected and amplified signals are obtained.

115- What is a Biorecognition layer on a biosensor? Give its components?

1. Biorecognition layer or Biosensing layer is platform for the Biosensing event in a biosensor.

2. Biosensing layer holds the Biological component of the Biosensor that is a Bioreceptor.
3. It is placed on the surface of a transducer in a Biosensor.
4. Biosensing layer determines the performance of Biosensor.

Basic Components of a Biorecognition layer:

There are **three main components** of the Biorecognition layer.

1. **Biorecognition element/Bioreceptor** element it could be any biological molecule having **strong affinity for its analyte**.
2. **A coating material** could be any polymer, gel or lipid
3. **A Binder** synthetic or natural

116- What is the impact of thickness on the performance on Biosensor?

1. **Thickness of the Biorecognition layer is very important** for the good performance of the Biosensor.
2. If the sensing **layer is too thin the process of Biorecognition** may not be occur properly.
3. **Too thick layer** may lead to **masking of the biorecognition signals**, as the distance between the bioreceptor element **analyte** complex and the transducer will be increased.

117- What are the forces that are involved in the non-covalent immobilization?

Non covalent immobilization is one of the methods of choice for attaching the biological molecule on the surface of transducer. There are two types of non-covalent physical methods.

Use of physical forces for non-covalent immobilization;

- **Hydrophilic interactions**
- **Hydrophobic interactions**

118- Amino Resins:

Amino resin is class of thermosetting resins formed by the interaction of

1. Amine, as urea, and
2. Aldehyde, as formaldehyde:

There are two types of amino resins:

- Urea-formaldehyde (UF)**
- Melamine-formaldehyde (MF).**

Both are used mainly as adhesives in the wood paneling industry.

119- How does Urea formaldehyde formed?

Urea formaldehyde (UF) resins are primarily made up of urea and formaldehyde with formaldehyde acting as the crosslinker. The UF resins are formed in water at a pH above 7.

Two steps of reaction

1. Formation of the methylol derivatives
2. Condensation

120- Epoxy Resins

Epoxy activated resins are ideal matrices to perform easy immobilization of enzymes/BRE.

1. They allow multipoint covalent binding between the enzyme and resin.
2. Epoxy methacrylate resins are epoxy activated resins produced with

I. High degree of crosslinking

II. Control porosity

121- Name the test formats that are used in Biosensors?

There are three types of methods that are established in laboratories are used in Biosensor technology.

1. Use of molecular switches:

The use of special class of molecules that can exist in more than one stable state with the change in the physical.

2. Immunological methods:

That method is based on antibody and antigen complex formation on a matrix.

122- Molecular switches:

The molecules may be shifted between the states in response to changes

1. PH
2. Light
3. Temperature
4. Electric current
5. Microenvironment
6. Ions and other ligands
7. Combination of stimuli

123- What is a DNA molecular switch?

1. DNA-based molecular switches are devices.
2. They are highly specific sensors, diagnostic tools.
3. DNA molecular switches, basic class of DNA nanostructures, turn some process.

- on
- off

depending on whether a substrate binds to the switch.

124- Three way Junction DNA switch

A series of three-way junction DNA-based molecular switches. It has two domains

1. **Recognition domain:** which interact with the target of interest and induces a structural change in the TWJ.
2. **Transduction domain:** which then becomes accessible and binds to a fluorescent molecule.

125- Give different types of ELISA formats that are used in diagnostic Biosensors?

Types:

Enzyme linked immune sorbent Assays (ELISA) assay based are one of the formats applied in biosensor measurements. In principle, four categories of test formats are available;

1. **Direct Detection**
2. **Sandwich assay**
3. **Competitive assay**

Direct Assay:

In direct assay, a **label-free transduction principle** is used.

1. It is a one-step process without any other reagents.
2. It allows **real-time monitoring** of **binding events** to the biosensor surface.

The signal response changes as soon as the analyte binds to the biorecognition element immobilized on the transducer surface.

Sandwich Assays:

In a sandwich there are **two bio recognition elements**.

1. **Capture antibody** that captures the target molecule.

2. **Detection antibody** that detects the capture antibody.
3. Response is obtained when the **detection antibody binds with the capture antibody**.

Competitive Assay:

In competitive Assay there are two types of Analyte.

1. In a competitive assay analyte and a known concentration of **labeled analyte** (derivative).
 2. The competitive assay could of two types
 - **Surface competitive assay**
 - **Solution competitive Assay**
-

Current pprz of Bt505

1- Write the types of controller system? 2 marks

Types of Controller system:

There are two types of controller systems in LOC.

1. **Controller system on Chip :**
 - All the necessary components are integrating all on-chip.
2. **Controller system off Chip :**
 - I. There is a **main system** to **control** the functioning of the chip.
 - II. The control panel is **not integrated** into the chip.

2- Name of four bioactive molecules use in biosenseing layers? 2 marks

Bioactive molecules:

1. Hydrogen peroxidase enzyme
2. Fluorescent Dye
3. Biotin
4. Streptavidin

3- Micro fluidics? 2M

The study behavior of fluids at micro level 100 Nano liters to 500 microliters is known as Micro fluids. A number of factors play important role in micro fluids.

- 1. Surface tension**
- 2. Energy dissipation,**
- 3. Fluidic resistance**

Due to flow of blood in capillaries, microfluidics is of great importance.

4- How many binding sites of Acetylcholine esterase? Write their names?

Acetylcholine esterase is a serine hydrolyzing enzyme. The basic structure of the AChE molecule is ellipsoidal. It possesses three binding sites, i.e.

1. Active site
2. Aromatic gorge,
3. Peripheral **anionic site, where the inhibiting compounds interact.**

5- Method of nanoparticles delivers into single cell of bio-analysis? 3 marks

Delivery Methods:

There are five methods for the delivery of optical sensors in the cell.

1. Endocytosis (non-specific or receptor-mediated);
2. Pico injection;
3. Liposomal delivery
4. **Membrane penetration peptide** can have a composition of positively charged amino acids
5. Gene gun

6- What is gravimetric analysis? What are gravimetric biosensors? 3marks

Gravimetric Transduction:

Gravimetric analysis is a **class of lab techniques** used to determine **the mass and Concentration of a substance** by measuring a **change in mass**.

1. The first **gravimetric biosensor** was a **quartz crystal microbalance (QCM)**.
2. It was based immune-sensor for protein detection, it was reported in 1972.

7- How does urea formaldehyde formed? 3 marks

Urea formaldehyde (UF) resins are primarily made up of urea and formaldehyde with formaldehyde acting as the crosslinker. The UF resins are formed in water at a pH above 7.

Two steps of reaction

1. Formation of the methylol derivatives
2. Condensation

8- Flourophore and quencher? 3 marks

1. **Fluorophore** is attached at one end of one arm and Quencher is attached at the end of other arm.

2. **Quenching** is a process which decreases the fluorescence intensity of a substance.

3. When the **fluorophore and quencher** are in **close proximity**, the quencher absorbs the energy emitted from an excited fluorophore, thereby **suppressing** its **emission**.

9- Biomarker characteristics? 3 marks

Certain universal characteristics are important for any biomarker such as;

- 1- they should be **non-invasive, easily measured, inexpensive, and produce rapid results**
- 2- they should be from readily available sources, such as blood or urine
- 3- they should have a high sensitivity, allowing early detection, and no overlap in values between diseased patients and healthy controls
- 4- they should have a high specificity, being greatly up regulated (or down regulated) specifically in the diseased samples and unaffected by comorbid conditions
- 5- biomarker levels should vary rapidly in response to treatment

10- Hexokinase enzyme importance? 3 Marks

Hexokinase **activates glycolysis by phosphorylating glucose**. Since the phosphorylation of glucose to glucose-6-phosphate is the rate limiting step of glucose metabolism, hexokinase has a very important role in regulating healthy glucose levels in the human body.

11- What is damage DNA? 3Marks

1. DNA can be **damaged** by **oxidizing agents, alkylating agents** and **high-energy electromagnetic radiation** such as **ultraviolet light** and **X-rays**.
2. DNA is damaged due to excessive levels of **Reactive Oxygen Species**. Normal level of ROS may be necessary for **memory and learning**.
3. DNA damaged is related to **cancer** and other diseases.

12- Ion selected electrode? 3Marks

An ion-selective electrode (ISE), also known as a specific ion electrode (SIE), is a **transducer (or sensor) that converts the activity of a specific ion dissolved in a solution into an electrical potential**. The voltage is theoretically dependent on the logarithm of the ionic activity, according to the Nernst equation.

13- How micro fluids maintain equilibrium on paper? 3 marks

When a droplet of liquid contacts with paper, the competition of **4 main phenomena**

1. Droplet wets the surface of paper
2. Paper absorbs and wicks the droplet by capillary action
3. Evaporation proceeds
4. Diffusion occurs.

All these four steps lead equilibrium conditions.

14- Features of lab on a chip? 3marks

- 1- **Safe Platforms:** The LOC platforms use **very minute amount of samples** and reagents, there is no need for **any proper** training and no specific safety precautions are required.
- 2- **Lower fabrication cost:** It allows mass production for cost effective **disposable chips**.
- 3- **Quality control:** The quality of the result is automatically verified.

15- Write the features of glassy carbon? 5 marks

1. High-**purity**
2. Excellent **Stability** as high as at **3,000 °C** in vacuum
3. Stable at **500 °C** in the air
4. **Inert** against Chemical **erosion**

5. **Impermeability** to Gas and Solution
6. Significant **Hardness**/ Strength
7. Favorable **electric conducting** property
8. Highly **Resistant** against **Inorganic and Organic salts**
9. Nonporous
10. Good **Bio-Compatibility**

16- What are the structure and functions of vitamin B12? What happens to body when it deficiency occurs? 5 marks

Vitamin B12:

1. The structure of B12 is based on a **corrin ring**
2. The central metal ion is cobalt.
3. There are six coordination sites.
4. Four coordination sites are provided by the corrin ring,
5. **One coordination site** is provided by a **dimethyl-benzimidazole group**.

Functions and Deficiency:

Vitamin B12 performs a key role in the synthesis of DNA, neurolipids, and cellular growth. Its deficiency causes.

1. Anemia
2. Confusion
3. Depression
4. Problems with balance
5. Poor memory

6. Soreness of the mouth or tongue

7. Tiredness

17- How trypsin enzyme activity stabilized? 5 marks

Trypsin stabilization:

Trypsin enzyme activity was stabilized in the presence of Trehalose and was immobilized by using two different techniques. Spray dried immobilization Freeze dried Immobilization.

In each case trehalose act as a stabilizer by forming **hydrogen bond with the enzyme** and the **glass matrix** hat was used for immobilization.

18- What is biochemical oxygen demand BOD?How you can measure BOD in industrial area? 5 M

Biochemical oxygen demand (BOD) represents **the amount of oxygen consumed by bacteria and other microorganisms** while they decompose organic matter under aerobic (oxygen is present) conditions at a specified temperature.

BOD measurement requires taking two samples at each site. One is **tested immediately for dissolved oxygen**, and the second is incubated in the dark at 20 C for 5 days and then tested for the amount of dissolved oxygen remaining.

19- Stick packs? 5M

Stick Packs:

1. **Tubular bags**, so-called stick packs, are widely used in the packaging industry.

2. It is used in LOC **pressure driven** devices.

20- What is the Test Format for biosensor? 5M

Types of Test Formats in Biosensor:

There are three types of methods that are established in laboratories and are used in Biosensor technology.

1. Use of molecular switches:

The use of special class of molecules that can exist in more than one stable states with the change in the physical.

2. Immunological methods:

Are based on antibody and antigen complex formation on a matrix.

21- Positive and negative photoresistor application? 5M

A photoresistor is a type of resistor whose **resistance decreases when the intensity of light increases**.

i- Positive photoresistors:

On the exposure of light positive photo resistor are dissolved.

ii- Negative photo resisters:

These photoresistor are not dissolved on exposure to light.

Example: Epoxy based polymers.

21- Architecture types on LOCs? 5M

There are different types of architectural design are available for LOC platforms.

1. Lateral flow platform

2. Linear actuated platform

3. Microfluidic large scale integration platform

4. Centrifuged microfluidics platform

5. Electro-wetting platform

22- Three way junction DNA is based on. Explain its domains? 5M

A series of three-ways junction DNA-based molecular switches. It has two domains

1. **Recognition domain:** which interact with the target of interest and induces a structural change in the TWJ.
2. **Transduction domain:** which then becomes accessible and binds to a fluorescent molecule.



Correct if you find any mistake